

Rochester Institute of Technology
Department of Computer Engineering
CMPE-630 Digital IC Design
Spring 2025
Laboratory Exercise 6
Design of a Mirror Adder

Part 1 Due: 03/07/2025
Part 2 Due: 03/21/2025, after spring break

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1 Introduction

The goal of this exercise is for students to learn how to design substantial CMOS circuits with proper sizing for equal drive strength. Additionally, students will be implementing new design concepts such as folding transistor for the layout design portion of the exercise.

Check section 4 for the lab report requirement and section 5 for the required signoff items. **Submit lab report (with the signoff sheet attached) to myCourses**

Keep the following in mind as you work through this lab.

- **NEVER** COPY CADENCE LIBRARY FILES FROM THE COMMAND LINE OR FILE MANAGER
- IF ANY OF YOUR CADENCE FILES ARE LOCKED, CLOSE THE TOOL AND EXECUTE THE FOLLOWING COMMAND:

```
/classes/ee620/bin/unlock_cadence.csh
```

- **Hotkeys:** You should learn how to use the hotkeys for better productivity. You can access a list of hotkeys (also known as bindkey) from CIW (the first window that pops up after you launch virtuoso). In CIW, select the menu option:

Options → Bindkey

Select a category (e.g. **layout**) and a list of current hotkeys (bindkeys) will be shown for it.

REMINDERS:

- Always start the Cadence tools using the **virtuoso &** command from your cmpe630 directory and NOT from your home directory.
- Always properly close the Cadence tools before you log off, otherwise you WILL lose your work and/or potentially lock your cells and not be able to access them!
- DO NOT delete any Cadence files from command line or file manager!

2 Procedure

2.1 Part 1

Create a brand new cell called "(FML)_MIRRORADDER" in your library and create a new schematic. *Note: Replace (FML) with your First, Middle, and Last initials.*

Create a new schematic view, and you will implement the schematic shown in Figure 1.

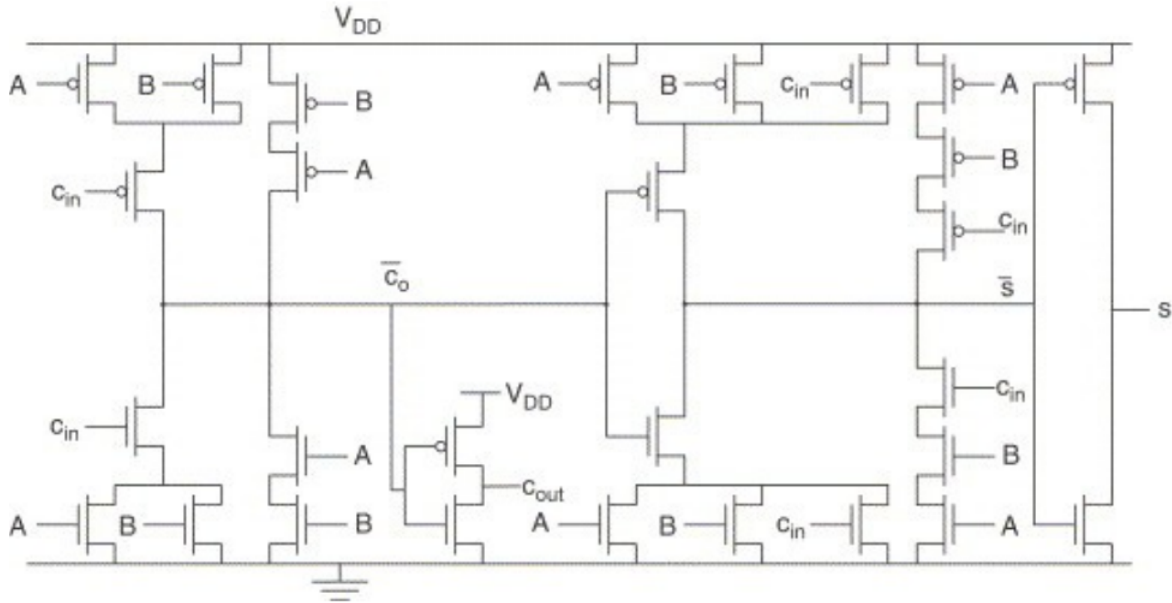


Figure 1: Mirror Adder Schematic

Remember to size transistor for equal pull-up and pull-down drive strength. Keep in mind that the minimum width for the 45nm technology is 120nm. The unit inverter aspect ratios specific to this technology are:

$$\frac{W}{L}_{NMOS} = 120/45 \quad (1)$$

$$\frac{W}{L}_{PMOS} = 240/45 \quad (2)$$

When you finish placing your transistors and connecting them. When setting input pins you should NOT be putting multiple pins! Instead, create necessary input pins at the top corner (A, B, Cin) then connect a little loose wire. Now, press "L" on your keyboard to create labels. You will label these as their names A, B, Cin. Then add little wires on each transistor and label them to their corresponding matching name. A small example is shown in the following Figure 2.

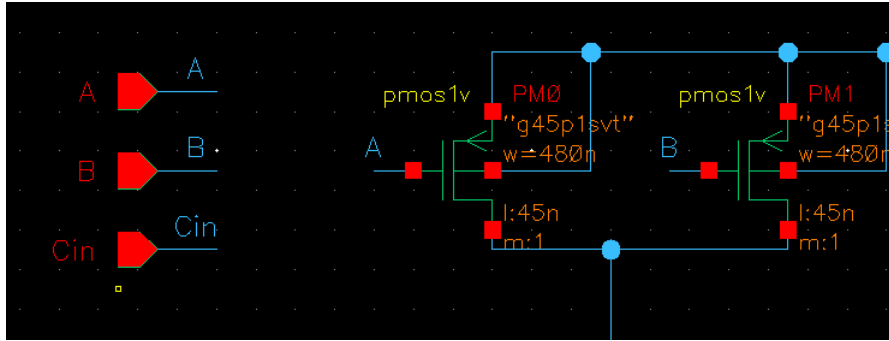


Figure 2: Adding Pins With Labels

Making a symbol. Ignore any warnings regarding VDD and VSS (We intentionally avoid them so the symbol is loose and ready to be used in a circuit for testing).

Optional: Once that is done, create the Verilog file and add the following code.

NOTE: You may receive a warning message window indicating that your symbol and Verilog module have pin mismatches. This warning can be safely ignored. DO NOT regenerate your symbol to fix the noted pin issues. (The mismatch occurred due to VDD and VSS pins being intentionally not added to your symbol; they do however appear in your Verilog module and schematic. We keep it this way so you can safely use your symbol in other schematics without any VDD and VSS attached.)

```
//Verilog HDL for "elm_lib", "ELM_MIRRORADDER" "functional"

module ELM_MIRRORADDER ( Cout, Sum, A, B, Cin, .VDD(\VDD! ), .VSS(\VSS! ) );

    input A;
    input B;
    input Cin;
    output Cout;
    output Sum;

    (* integer inh_conn_prop_name = "VDD";
       integer inh_conn_def_value = "cds_globals.\VDD! "; *)
    \VDD! ;

    (* integer inh_conn_prop_name = "VSS";
       integer inh_conn_def_value = "cds_globals.\VSS! "; *)
    \VSS! ;

    assign signal_1 = A ^ B;
    assign signal_2 = signal_1 & Cin;
    assign signal_3 = A & B;

    assign Sum = signal_1 ^ Cin;
    assign Cout = signal_2 | signal_3;

endmodule
```

Figure 3: 1-bit Mirroradder Verilog Code

Follow up by making a testbench (please refer to Lab2 if you need a refresher on creating ADE XL tests). This time you will add two variable capacitors, both with capacitance CAP and you will connect one in each output.

Use ADE XL and create simulation tests and timing simulations of the Mirror Adder. The following timing measurement tests you will perform are the ones listed below. You will have to do these with **no load** and the same ones **with load of 120fF**. **Temp = 125C** and **VDD = 1.0V** are used for all cases. Additionally, you will also need to perform **a transient simulation** to ensure your mirror adder functions as intended.

You will measure Rise/Falltime and propagation delay (high to low or low to high) with EACH Test: There will be 32 measurements for unloaded and 32 for loaded (you may use a design variable to change the capacitance without duplicating tests). **You must label measurements/tests with descriptive names. You must follow the same naming order as the tests shown below.** *Note: You will have a total of 64 measurements!! Good luck and be patient*

- Carry Rising (measure Rise Time and T_{PLH} for each one below):

A 0 → 1, B = 1, C = 0

A = 1, B 0 → 1, C = 0

A = 1, B = 0, C 0 → 1

A = 0, B = 1, C 0 → 1

- Carry Falling:

A 1 → 0, B = 0, C = 1

A = 0, B 1 → 0, C = 1

A = 0, B = 1, C 1 → 0

A = 1, B = 0, C 1 → 0

- Sum Rising:

A 1 → 0, B = 0, C = 1

A = 0, B 1 → 0, C = 1

A = 0, B = 1, C 1 → 0

A = 1, B = 1, C 0 → 1

- Sum Falling:

A 0 → 1, B = 1, C = 0

A = 1, B 0 → 1, C = 0

A = 1, B = 0, C 0 → 1

A = 1, B = 1, C 1 → 0

2.2 Part 2 Prelab

The three transistors that measure 720nm are too large to fit in the PMOS portion of the cell since we have a height constraint. To accommodate the design with this constraint, we have to convert these transistors into folding transistors. If you recall from lecture, a transistor can be split in half of it's size in more than two as long as they are placed in parallel.

Redraw your schematic with folding transistors (NOT on your actual cadence schematic). Then, **create the Euler's path and draw your stick diagram.** Please use colors! Since this is a big design, having a color-coded blueprint will be an immense help when you start working on your layout design.

You must finish those two before coming to the second week of Lab 6! The stick diagram needs to be detailed, colored, probably labeled, and not impossible to read

2.3 Part 2 In-Lab

When starting this lab, make sure you are on a fresh logged in session. If not, then close Cadence, log out and log back in. Open the Cadence tools. Now create the layout view for the FML_MIRRORADDER. From the Library Manager window, select your library, then your Mirror Adder cell. Then select **File** → **New** → **From Cellview** .

Then, create the layout based of your stick diagram. Be mindful of the design rules and try your best to use the minimum measurements since you have to fit in a lot of stuff in a constrained space (height wise). Remember that you can change the width as you go along by clicking on your template cell and pressing 'Q' on your keyboard.

YOU MUST FOLLOW THE LAYOUT DESIGN GUIDELINES PROVIDED IN LAB5! Please refer to Lab5 for those guidelines. As mentioned there, we will not be able to provide signoff if your adder layout does not meet all of the guidelines and rules.

Things to keep in mind how folding transistors work and be mindful of the double sources and double drains and appropriately connect them.

Test DRC, LVS and RCX. Once your design is done, use the same testing convention shown in Lab Exercise 4 to perform a post-layout test then fill in the results tables for Pre-layout and Post-Layout portion (see Table 1 and 2).

There is an extra credit in this lab involving combining 4 1-bit mirror adders into a 4-bit mirror adder. For those who want to attempt that, we recommend designing your layout so that Cin pin is placed on one edge of the layout and Cout is placed on the other edge.

2.4 Extra Credit (You must not use Metal2 for routing!)

This *might* be helpful for the final project. In this portion of the lab, you will be designing a 4-bit ripple carry adder using the 1-bit mirror adder schematic and layout. Create a brand new cell schematic. The name will be **FML_MIRRORADDER_4BIT**.

In your schematic **Create** → **Instance**. Then click Browse and look for your mirror adder symbol like this:

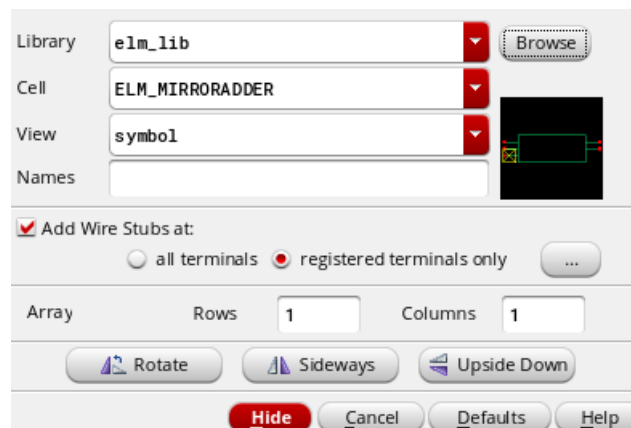


Figure 4: New Cell Form

Then, create three more to have four instances of the mirror adder and connect them. You should have a total of four instances connected to make the ripple carry adder.

This schematic will be different from previous ones since we are dealing with multiple bits. The inputs will be buses. Add little pieces of wires to the outputs without pins and label each ' $A < 0 >$ ' then ' $B < 0 >$ '. An example is shown below.



Figure 5: Labeling example

Follow the rest of the inputs for each instance as $A < 1 >$, $B < 1 >$ etc, all the way to bit 3. Then separately, create input pins for the bus. The name for the pin **has to be** ' $A < 3 : 0 >$ ' and ' $B < 3 : 0 >$ ' then C_{in} stays normal since it's just a 1 bit input. Then add a Wide piece of wire with labels of the same name. Your final schematic should look similar to the following. **Do not add VDD or VSS in this schematic.**

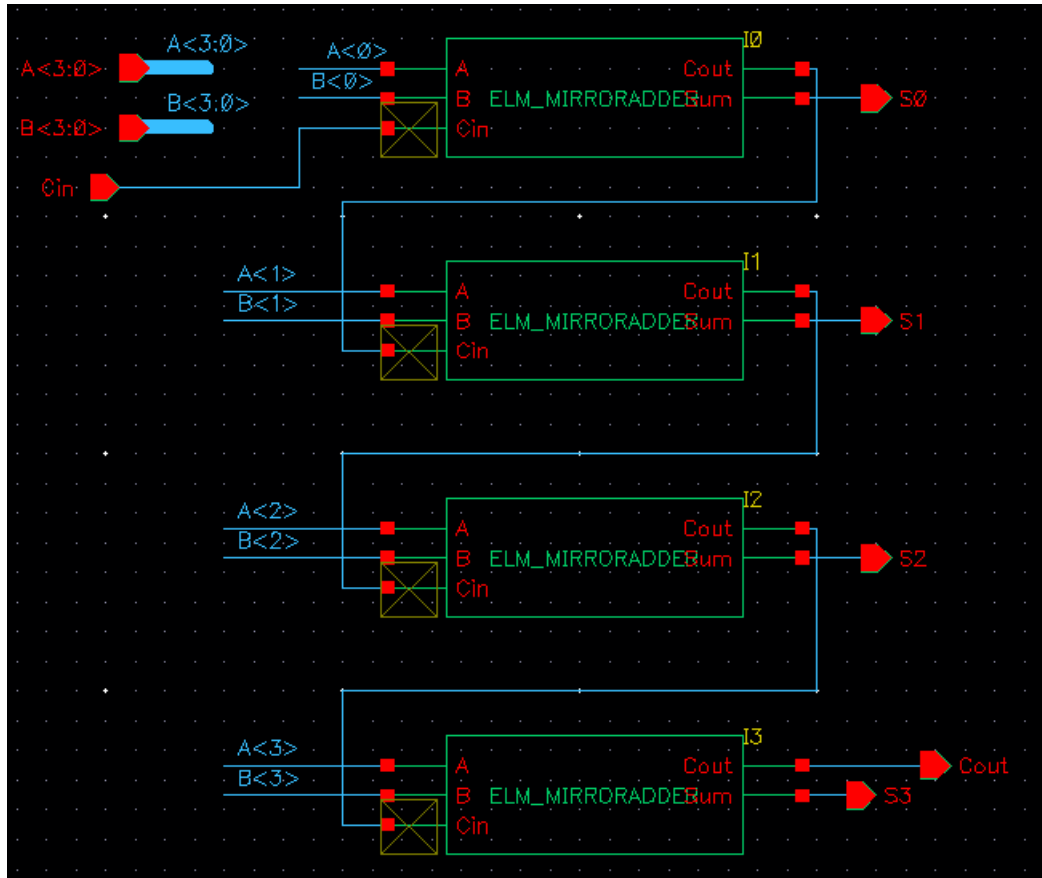


Figure 6: Four-bit Schematic

Check and save your work. Then, create a symbol for this schematic. Follow the same instructions for the simulation process. In your testbench schematic you will have to use an instance with generated Verilog code. Go to **Create** → **Instance** → **ahdlLib** and find **code_gen_4bit**, place two of these for the A and B inputs. Then click on it and press 'Q' and set up the following parameters shown below.

Apply To	only current ☒ instance ☒	
Show	☐ system ☒ user ☒ CDF	
Browse Reset Instance Labels Display		
Property	Value	Display
Library Name	ahdlLib	off ☒
Cell Name	code_gen_4bit	off ☒
View Name	symbol	off ☒
Instance Name	I2	off ☒
Add Delete Modify		
CDF Parameter of view	Use Tools Filter ☒	Display
tperiod	20n	off ☒
vlogic_high	1	off ☒
vlogic_low	0	off ☒
tdel	0	off ☒
trise	100p	off ☒
tfall	100p	off ☒
model		off ☒

Figure 7: Veriloga Code Gen Parameters

You should add a capacitor variable CAP at the end of all your outputs S0 to S3 and Cout. Your testbench schematic should look like Figure 8. Note that the inherit instances of VDD and VSS are connected to output pins with the instance names as followed (You do not have to label the VDD and VSS wires). However, do not forget to label your wires like shown in the example. **These labels are important, otherwise your design will fail.**

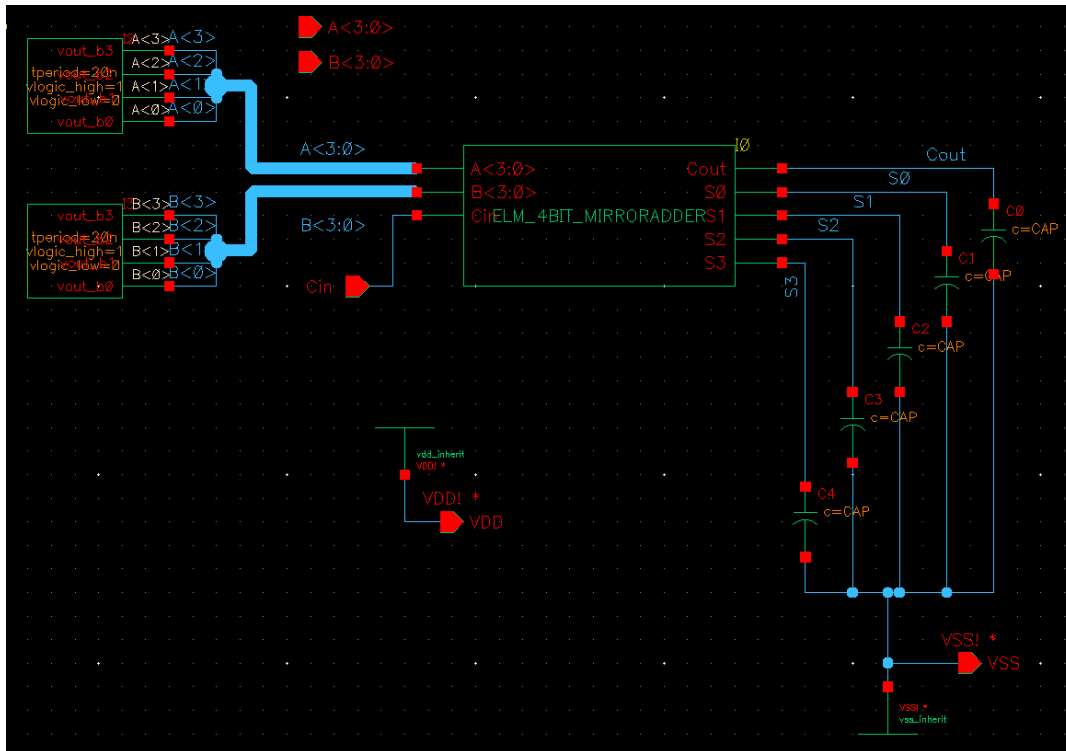


Figure 8: Four-bit Testbench Schematic

Proceed to following the instructions from the previous tutorials on two to create tests. This test will require slightly different changes in the test editor's stimuli forms. Make sure they match with the following.

Figure 9: Inputs Form

Figure 10: Global Sources Form

Make sure your capacitor variable CAP is set to 100f and that is checked off in the Global Variables section of your Data View. Once everything is set up, run the tests and plot them. Then measure the waveforms and fill out the tables from the Results section for the pre-layout portion.

Afterwards, create a layout cell for your four-bit mirror adder. Make sure the layout settings match to those instructed in Lab Exercise 4. Then add 4 instances of your 1-bit mirror adder layout. Connect them to make a 4 bit ripple carry adder; use Metal 1 to connect all VDD, VSS, then route your Cout to Cin to connect the rest of the layout together also using Metal 1.

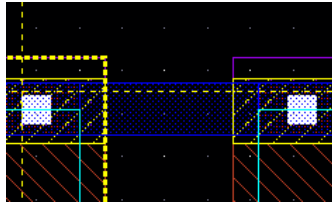


Figure 11: Metal1 Connection Between VDDs Example

Test DRC, LVS and RCX. Once your design is done. Performing a transient simulation to show that your design works functionally as a 4-bit adder. Your simulation should be able to perform the following test cases.

- $A = 4'b0000$, $B = 4'b0000$, $C_{in} = 1'b0$
- $A = 4'b1111$, $B = 4'b0000$, $C_{in} = 1'b1$
- $A = 4'b0000$, $B = 4'b1111$, $C_{in} = 1'b0$
- $A = 4'b1100$, $B = 4'b0010$, $C_{in} = 1'b1$
- $A = 4'b0001$, $B = 4'b1100$, $C_{in} = 1'b0$
- $A = 4'b1111$, $B = 4'b1111$, $C_{in} = 1'b1$

2.5 Results

From all the data that you gathered. Pick the worst-case timings (the largest value) and place them on these tables. **You will have two of each one of these tables as you will be recording for pre-layout and post .**

Note: The completed tables are required in the lab report.

Table 1: Sum Results

	Vdd (V)	Temp (C)	Load (fF)	RiseTime	FallTime	T_P,HL	T_P,LH
Worst Case	1	125	0				
	1	125	120				

Table 2: Carry Out Results

	Vdd (V)	Temp (C)	Load (fF)	RiseTime	FallTime	T_P,HL	T_P,LH
Worst Case	1	125	0				
	1	125	120				

Table 3: Overall Results

	VDD (V)	Temp (C)	Load (fF)	$F_{input,max}$	$F_{throughput,max}$
Sum	1	125	0		
	1	125	120		
Cout	1	125	0		
	1	125	120		

$$F_{input,max} = \frac{1}{t_{rise} + t_{fall}} \quad (3)$$

$$F_{throughput,max} = \frac{1}{T_{P,HL} + T_{P,LH}} \quad (4)$$

3 Questions

1. Calculate theoretically (by hand) the worst case propagation delay for the sum output of the whole mirror adder schematic. Using the RC and elmore delay model.

4 Lab Exercise Report Requirements

This are the requirements for your lab report. Please try your best to use the right format for CE lab exercise reports you've been using these past many years you've been in the program. This is a 5th year grad level course and you should be equipped with a lot of knowledge to write a proper report for this class. Please follow the list guidelines and also apply as much as you can from what you already know on how to explain theory, waveforms, results, behavior and everything else that should be inside a proper report!

Note: Proper citation is needed for all non-original figures (i.e., those that you have not made yourself).

1. Cover Page

- Name and number of the lab exercise
- Your name, name of the instructor and TA
- Submission Date

2. Table of contents

3. Abstract

- A description of what you did for this exercise, how you did it, and significant numerical lab results

4. Design Methodology

• PART 1

- Provide a theoretical discussion about transistor sizing
- Describe the motivation of your design, e.g. why is it good, necessary or important
- Have a brief discussion on the properties and the logic function (truth table) of the mirror adder properties
- Capture and include the mirror adder schematic with proper sizing
- Discuss your theoretical approach to minimizing delays and its limitations
- Include testbench schematic

• PART 2

- Describe the motivation of your design, e.g. why is it good, necessary or important
- Include the stick diagram of your mirror adder (use colors and must be properly labeled)
- Discuss the Euler's path of the mirror adder
- Discuss what folding transistor is and how it applies to the mirror adder layout
- Discuss your theoretical approach to minimizing delays and its limitations

5. Result and Analysis

- Include the following simulation results:
 - Transient simulation waveform of the mirror adder
 - Pre-extraction test results in the ADE XL editor

– Post-extraction test results in the ADE XL editor

- Include clean DRC and LVS output message
- Include a capture of your **Extracted Parasitic Layout** and briefly discuss it
- Complete and include all of the tables listed in section 2.5. The tables must be properly presented and labeled (do not forget units!)
- Calculate percentage increase or decrease in the maximum operating frequency and the maximum throughput from the loaded case to the unloaded case
- Report the width, height, and area of your layout(s) (1 bit mirror adder and 4 bit if you completed extra credit)

6. Conclusions

- Restate important results, how they were obtained, any uncertainties, possible improvements, and any implications
- This should not be a copy-paste of the Abstract

7. Questions

- Re-write the questions(s)
- Answer all of the questions from the “questions” section.

8. Reference (if needed)

9. Sign-off Sheet

- **This is absolutely required! Lab reports will not be graded without a sign-off sheet, resulting in an automatic zero.**

5 Signoff Sheet

Table 4: Design of a Mirror Adder

Students Name:

Section:

Part 1 Demo	Point Value	Points Earned	Date
Schematic Sizing Check	5		
Schematic CIW Check	10		
Pre-layout Test Simulation Demo	20		
Part 2 Pre-Lab	Point Value	Points Earned	Date
Schematic with Folding Transistors	5		
Mirror Adder Stick Diagram	5		
Part 2 Demo	Point Value	Points Earned	Date
DRC Check	15		
LVS Check	15		
Parasitic RC RCX	10		
Post-layout Test Simulation Demo	15		
Extra Credit 4-bit reqs + report	+10 overall		
Total Points	100		